

Email Spam Classifier

PROJECT DESCRIPTION:

In the age of digital communication, spam emails pose a serious threat by flooding inboxes with irrelevant or malicious content. This project aims to build a **Machine Learning-based Email Spam Classifier** that can automatically distinguish between **spam and non-spam (ham)** emails by analyzing the email content.

The system is trained on a labeled dataset of emails using natural language processing (NLP) techniques and supervised learning algorithms. It learns to recognize common patterns in spam messages, such as specific keywords, formatting styles, and sender behavior, enabling accurate prediction for new incoming emails.

☀️ FEATURES:

- Detects whether an email is spam or legitimate (ham)
- Uses text processing to extract meaningful features from emails
- Lightweight and fast — can be integrated into existing email systems
- Real-time classification of emails
- High accuracy with low false-positive rate
- Option to visualize important spam-indicating keywords

🧰 TECHNOLOGIES USED:

- **Language:** Python
- **Libraries:**
 - scikit-learn – for model training and evaluation
 - pandas, NumPy – for data handling
 - NLTK or spaCy – for natural language processing

- Matplotlib / Seaborn – for visualizations
- **Algorithms Used:** Naive Bayes, Logistic Regression, Support Vector Machine (SVM)
- **Dataset:**
 - **SpamAssassin Dataset** or **UCI SMS Spam Collection Dataset**
 - Kaggle Email Spam Datasets :
<https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset>

OUTCOMES:

- Successfully trained and validated ML models to classify emails as spam or ham
- Achieved high accuracy and precision using NLP + ML techniques
- Developed a scalable prototype for spam filtering
- Strengthened understanding of text classification and data preprocessing
- Can be extended to detect phishing emails, scams, or ads